XUEQING WU

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EDUCATION

University of California, Los Angeles 09/2023 - Present
PhD in Computer Science (in progress). Advisor: Kai-Wei Chang, Nanyun Peng.

University of Illinois Urbana-Champaign 08/2021 - 05/2023
MS in Computer Science. Advisor: Heng Ji.

University of Science and Technology of China 09/2016 - 06/2020
BS in Electronic Engineering and Information Science. GPA: 4.03/4.30 (rank: 1/363).

RESEARCH EXPERIENCE

University of California, Los Angeles 09/2023 - Present
Research Assistant. Mentor: Kai-Wei Chang, Nanyun Peng

- Proposed Planned Test-Time Scaling (PTTS), a planner-executor framework that coordinates diverse reasoning paths instead of independent repeated sampling. Developed zero-shot and RL variants that improve pass@64 by up to 13.4 points across five math reasoning benchmarks. **(Preprint)**
- Investigated *asymmetric optimization* in VLM post-training, where reasoning improves disproportionately over perception. Diagnosed distinct causes in SFT and RL and developed targeted mitigation recipes, boosting accuracy by up to 18.2 and 6.0 points, respectively. **(Preprint)**
- Established VISCO, a novel benchmark for evaluating critique and correction capabilities of VLMs, two critical skills towards VLM *self-improvement*. Evaluated 24 VLMs and identified critique as a key bottleneck. Proposed LookBack method that improves performance by up to 13.5%. [**CVPR 2025**]
- Proposed VDebugger, a critic-refiner framework that debugs visual programs by reasoning over execution feedback, pinpoints error-inducing location and corrects the input program. Improved accuracy by up to 3.2% on six visual reasoning benchmarks. [**EMNLP Findings 2024**]

University of Illinois Urbana-Champaign 08/2021 - 05/2023
Research Assistant. Mentor: Heng Ji

- Established a better benchmark for *open-vocabulary state tracking* with high-quality human-annotated dataset and robust evaluation metrics. Proposed two techniques, *entity memory* and *entity-conditioned prediction*, to improve the performance. [**ACL Findings (short paper) 2023**]
- Proposed a novel task of *cross-document misinformation detection*, including document-level and fine-grained *event-level* detection. Designed a GNN-based detector that reasons over cross-document knowledge graph and significantly outperforms existing methods by up to 7 F1. [**NAACL 2022**]

INTERNSHIP EXPERIENCE

Google 03/2026 - 09/2026
Student Researcher. Mentor: Jinsung Yoon *Sunnyvale, CA*

- Established a benchmark for multi-turn scientific diagram refinement with simulated user feedback, and identified quality drift and forgetting as key limitations. Proposed a multi-agent refiner that improves quality by up to 18.6 points and reduces forgetting by up to 6.2 points. **(Preprint)**

Meta Superintelligence Labs 06/2025 - 09/2025
Research Intern. Mentor: Yeming Wen *Menlo Park, CA*

- Benchmarked multi-turn, multi-modal front-end coding with a novel agent-based evaluation framework, and identified forgetting and visual interpretation gap as key limitations. Proposed *agent-based self-critique* that reduces forgetting to near zero and enhances performance by up to 9.5%. **[CoLM 2026]**

ByteDance AI Lab 05/2023 - 09/2023
Research Intern. Mentor: Haoran Huang *Shanghai, China*

- Proposed the novel task of tabular data analysis on complex, real-world end-user queries. Curated DACO dataset with 440 databases, ∼2k training data, and ∼200 manually annotated test data. Benchmarked various algorithms such as fine-grained RLHF. [**NeurIPS D&B 2024**]

IBM Research 05/2022 - 08/2022
Research Intern. Mentor: Alfio Gliozzo *Yorktown Heights, NY*

- Proposed a retrieval-augmented model for table augmentation tasks including cell filling, row population and column population. Achieved state-of-the-art on two datasets with absolute MRR gains of up to 30% compared to non-retrieval baselines. [**ACL Findings 2023**]

ByteDance AI Lab 07/2020 - 07/2021
Research Intern. Mentor: Hang Li *Beijing, China*

- Proposed and benchmarked *text-to-table*, a novel IE setting that extracts table-format information and requires no pre-defined schema. Adopted a seq2seq approach that significantly outperforms named entity extraction and relation extraction. [**ACL 2022**]

Microsoft Research Asia 10/2019 - 06/2020
Research Intern. Mentor: Tao Qin *Beijing, China*

- Proposed a novel sequence learning framework that boosts a given main task using auxiliary training tasks. Designed a novel RL algorithm to jointly train the base model and task scheduler, which improved the baselines on four simultaneous translation tasks and a stock forecasting task. [**ICML 2021**]

PREPRINTS & PUBLICATIONS

- Bingxuan Li, Yiwen Song, **Xueqing Wu**, Yanzhou Pan, Yang Li, Kuang Su, Jingyun Liu, Sebastian Ko, Huan Zhang, Tong Zhang, Nanyun Peng, Tomas Pfister, Yale Song, *VIS-Ground: Video Interactive Storytelling with Contextual Grounding*, Preprint. 2026. Link
- **Xueqing Wu***, Langxing Bai*, Hritik Bansal, Po-Nien Kung, Shuo Li, Hao Liu, Nanyun Peng, Kai-Wei Chang, *Planned Test-Time Scaling with Coordinated Reasoning Paths*, Preprint. 2026. Link
- **Xueqing Wu**, Ashwin Balasubramanian, Bingxuan Li, Dawei Zhu, Kai-Wei Chang, Yale Song, Yiwen Song, Rui Meng, Tomas Pfister, Nanyun Peng, *PaperBanana-Interact: Scientific Diagram Refinement with Multi-Turn Human Feedback*, Preprint. 2026. Link
- **Xueqing Wu**, Yu-Chi Lin, Kai-Wei Chang, Nanyun Peng, *On Asymmetric Optimization of Reasoning and Perception in Vision-Language Model Post-Training*, Preprint. 2026. Link
- **Xueqing Wu**, Zihan Xue, Da Yin, Shuyan Zhou, Kai-Wei Chang, Nanyun Peng, Yeming Wen, *FronTalk: Benchmarking Front-End Development as Conversational Code Generation with Multi-Modal Feedback*, CoLM. 2026. Link
- **Xueqing Wu***, Yuheng Ding*, Bingxuan Li, Pan Lu, Da Yin, Kai-Wei Chang, Nanyun Peng, *VISCO: Benchmarking Fine-Grained Critique and Correction Towards Self-Improvement in Visual Reasoning*, CVPR. 2025. Link

- **Xueqing Wu**, Zongyu Lin, Songyan Zhao, Te-Lin Wu, Pan Lu, Nanyun Peng, Kai-Wei Chang, *VDebugger: Harnessing Execution Feedback for Debugging Visual Programs*, EMNLP Findings. 2024. Link
- Zi-Yi Dou, Cheng-Fu Yang, **Xueqing Wu**, Kai-Wei Chang, Nanyun Peng, *Reflection-Reinforced Self-Training for Language Agents*, EMNLP. 2024. Link
- **Xueqing Wu**, Rui Zheng, Te-Lin Wu, Hanyu Zhou, Tang Mohan, Kai-Wei Chang, Nanyun Peng, Haoran Huang, *DACO: Towards Application-Driven and Comprehensive Data Analysis via Code Generation*, NeurIPS Dataset and Benchmark Track. 2024. Link
- Xiao Liu, Zirui Wu, **Xueqing Wu**, Pan Lu, Kai-Wei Chang, Yansong Feng, *Are LLMs Capable of Data-based Statistical and Causal Reasoning? Benchmarking Advanced Quantitative Reasoning with Data*, ACL Findings. 2024. Link
- Michael R. Glass, **Xueqing Wu**, Ankita Naik, Gaetano Rossiello, Alfio Gliozzo, *Retrieval-Based Transformer for Table Augmentation*, ACL Findings. 2023. Link
- **Xueqing Wu***, Sha Li*, Heng Ji, *OpenPI-C: A Better Benchmark and Stronger Baseline for Open-Vocabulary State Tracking*, ACL Findings (short paper). 2023. Link
- **Xueqing Wu**, Kung-Hsiang Huang, Yi Fung, Heng Ji, *Cross-document Misinformation Detection based on Event Graph Reasoning*, NAACL. 2022. Link
- **Xueqing Wu**, Jiacheng Zhang, Hang Li, *Text-to-Table: A New Way of Information Extraction*, ACL. 2022. Link
- **Xueqing Wu**, Lewen Wang, Yingce Xia, Weiqing Liu, Lijun Wu, Shufang Xie, Tao Qin, Tie-Yan Liu, *Temporally Correlated Task Scheduling for Sequence Learning*, ICML. 2021. Link
- **Xueqing Wu**, Yingce Xia, Jinhua Zhu, Lijun Wu, Shufang Xie, Tao Qin, *A Study of BERT for Context-Aware Neural Machine Translation*, ACML journal track. 2021. Link
- **Xueqing Wu**, Yingce Xia, Jinhua Zhu, Lijun Wu, Shufang Xie, Yang Fan, Tao Qin, *mixSeq: A Simple Data Augmentation Methodfor Neural Machine Translation*, IWSLT Workshop. 2021. Link
- Yixing Zhu, Jun Du, **Xueqing Wu**, *Adaptive Period Embedding for Representing Oriented Objects in Aerial Images*, IEEE Transactions on Geoscience and Remote Sensing. 2020. Link

ACADEMIC SERVICE

Conference Reviewer: ICLR (2026), ACL (2024–2026), NAACL (2024–2025), EMNLP (2024–2026), CVPR (2025), NLPCC (2024), SoCalNLP (2023)

AWARDS & HONORS

Amazon Fellow, University of California, Los Angeles	09/2025
Graduate Dean's Scholar Award, University of California, Los Angeles	09/2023
Siebel Scholar (awarded annually to <100 students from around the world)	09/2022
Guo Moruo Scholarship (top 1%), University of Science and Technology of China	10/2019
Tang Lixin Scholarship, University of Science and Technology of China	10/2018
China National Scholarship (top 0.2% of national candidates), China	10/2018